

Multi-modal Conversational Search for People with Intellectual Disability: An Exploratory Study

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ABSTRACT

Like us all, people with intellectual disability are self-motivated learning and are enthusiastic information seekers. However, there are some technical challenges for people with intellectual disability when accessing required information and interacting with conversational search systems. This paper presents the preliminary development of a simulated multi-modal conversational search system to understand its potentials and limitations for people with intellectual disability. To conduct an exploratory study, we developed a Wizard of Oz conversational multi-modal system which records user activities, including touch, text and verbal interactions. Then, we preliminarily evaluated the system with eight adults with intellectual disability in one of two different information-seeking scenarios – in an individual and collaborative setting. Drawing on the findings and insights of the study, participants valued the system as it fulfilled their information retrieval needs, and it achieved an acceptable level of accessibility. These findings and future research into this topic can guide accessibility improvements to current search systems and the development of future conversational search systems.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility**; • **Information systems** → **Information retrieval query processing**.

KEYWORDS

intellectual disability, conversational search, accessibility

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1 INTRODUCTION

As advancements in artificial intelligence technology are increasingly pervasive in several aspects of life, searching for information

has become easier than ever before. Basic conversational search can currently be observed in voice-activated assistants such as Siri, Alexa, and Google Home. These systems have seen high rates of adoption, due to their ability to perform everyday tasks faster [13]. These interfaces have also been notably useful for supporting agency and independence of people with intellectual disability, as they remove the barrier of spelling and typing associated with text-based information retrieval. As a result, a high satisfaction rate with voice-activated interfaces (VAIs) was found when interviewing people with intellectual disability, with 72% of participants preferring voice-activated interfaces over typing [1]. However, currently available conversational search systems lack multi-modal communication, as they do not always show images results, or allow communication through alternative methods such as touch. This has been noted as a potential accessibility barrier, as people with intellectual disability have been shown to be driven by images to assess the relevancy of results when information seeking [16]. Allowing alternative methods of communication is also important, as formulating queries can sometimes be difficult. These identified accessibility barriers of current VAIs suggest that conversational multi-modal search could potentially increase the accessibility of information. Several studies have investigated user experience when interacting with a simulated fully conversational search, however this has not been undertaken with people with intellectual disability. The severity of an intellectual disability varies across conditions and individuals. Accessible design has the potential to increase people's confidence in their own searching capabilities. Additionally, there is lack of research into how to combine multi-modal search with conversational search. In order to gain a better understanding of the requirements for conversational multi-modal search for people with intellectual disability, simulating multi-modal conversational search allows to log conversations between users and the experimenter for further analysis. This research could guide improvements in accessibility for existing voice assistants and guide the future development of conversational search. Detailed insight will be discussed into how people with intellectual disability seek information online without the limitations of commercial search engines, or the social aspect of conversing directly with a human for information.

2 BACKGROUND AND RELATED WORK

2.1 Information Retrieval Process for People with Intellectual Disability

The information retrieval process was described in behavioural user models, explaining the typical interactions which occur between a user and information in an iterative search process. The

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information-seeking framework [10] was proposed the sub-activities of; recognizing an information need, accepting the challenge to act, formulating the problem, expressing the need within the constraints of the interface, examining the results, reformulating, and using the information [11]. When this neurotypical-based behavioural model was compared to the information-seeking behaviour observed from people with intellectual disability, it was found to be suitable when an additional stage of leveraging social connections or assistive technologies was added [16]. The queries formulated for voice-activated interfaces were observed to be longer, with the inclusion of natural language usage of introductory terms and polite sign-offs [1]. Within the formulation and expression stage, Balasuriya et al. [1] observed that users indicated a higher satisfaction and independence when using voice compared to text-input, due to the removal of the barrier of spelling and typing. This is further supported by the cognitive demands required when writing, which can be challenging, making voice input potentially much faster and easier [12]. However, some participants preferred the text-based search, due to frustration caused by voice assistants misunderstanding speech impediments or not listening to full queries [1]. When examining the information returned by these systems, most participants focused on images rather than words/titles [17]. Images were shown to help supplement text to help people with intellectual disability better understand what it is about [2]. When searching using voice interfaces, people with intellectual disability found the voice answers returned beneficial, with a participant indicating Siri gave 'clear answers' as she did not have to read the results herself [1].

2.2 Multi-modal Information Retrieval

Image: Users with intellectual disability have been shown to struggle with text input and sometimes understanding textual information [3]. Images could serve as both an input for users to express their queries and as a response to communicate information. The potential for people with intellectual disability to express information with images was seen by Balasuriya et al. [2], where participants chose to answer comprehension questions correctly by pointing at images instead of verbalising answers. Keskinen et al. [5] observed similar results with SymbolChat messaging, where people with intellectual disability could express themselves in spontaneous communication through symbols, without any training. Particularly, images help to support textual information for people whose disability affects their reading comprehension and literacy ability [20].

Touch: The traditional way to interact with devices using mouse and arrow on screen is demanding. It was considered as a potential barrier for people with intellectual disability who experience impairments in motor skills or visual perception abilities [3]. Touch screen interfaces identified as a more accessible input method as it is easier to understand a cause-and-effect relationship between their actions [9]. Kumin et al. [8] also observed a high rate of success when 10 adults with down syndrome used multi-touch devices for multiple-step tasks, which often involved searching. Touch interfaces also have the potential to support communication due to their support of the intuitive communication method of pointing [15].

Voice Voice input search engines can help people with intellectual disability complete tasks faster and with less help [1]. Voice responses are also clearer than text for people who may struggle with literacy [1], but demanding on short-term memory for large amounts of information [19]. Seeking assistance is often a part of the information-seeking process for people with intellectual disability. Voice input has the potential to allow a more collaborative approach to information-seeking [17]. However, current voice interfaces do not support collaborative search, as they often cannot distinguish between multiple voices [7].

2.3 Conversational Search

Conversational search allows users to submit queries and receive answers or questions in the form of a natural conversation. A conversational system simulated on real-world user's datasets from e-commerce scenarios showed a higher rate of performance when the artificial intelligence was personalized with past purchase history and vocabulary preferences [21]. Memory in conversational systems can therefore be used to remember what was previously said by the user or system to assist in resolving future information needs more successfully [14]. The ability to hold memory of past and current conversation is a key advantage of conversational search, as maintaining context has been shown as essential for satisfaction when interacting with intelligent assistants [6].

There is, as yet, little research in the potential of combining multi-modal and conversational search, and the potential it holds for improving accessibility to online search. This exploratory study addresses this gap by identifying the opportunities of multi-modal input and outputs that match the different strengths of people with intellectual disability.

3 METHOD

We applied an ability centered approach from previous research [18] to engage and gain understanding of how people with intellectual disability use various modalities for information seeking, and how much context we should understand. We discuss the design of a Wizard-of-OZ platform, and how to set-up the accessible prototype before and throughout different settings/individuals.

3.1 Phase 1: Preparation

3.1.1 Design brief. The design is intended as a starting point, and we are not trying to validate any design principles, rather than investigating to include multi-element in this study. Our primary aim is to ensure the application is accessible and, at this stage, the design feature for the user we look at the accessibility. In the further stage, we can build many features and other design aspects.

For the user side, figure 1 left, the user interface is similar to the existing search website with a search bar at the top of the page and the information towards the bottom [4]. Cartoonish images and icons have also been shown to communicate the information well to people with ID [1]. Therefore, a robot image and pulsating microphone icon were also added to the top of the page to improve communication to the user of the option of voice input. As people with intellectual disability have been shown to prefer a flat search page over searching through nested links [4], this user interface will offer information in the form of image grids, text-to-speech

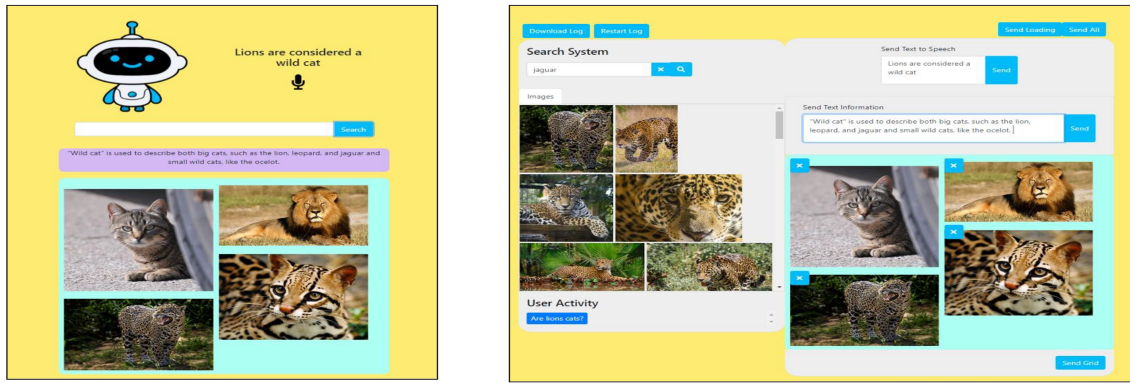


Figure 1: An initial prototype of simulated multi-modal conversational search system - on the left is the user view and on the right is the helper/experimenter view

voice, and text. Users will be able to respond or initiate queries with their voice, typical text search, or touching on the image grid. For the experimenter side, we used an iterative approach and Wizard of Oz is within our context. An initial experimenter page was developed with regular application deployment cycles with testing and feedback from other researchers. As recommended from VAI's study [1], the system's voice search should act as a constant live stream, with no time-outs or immediate failures from mispronunciation; this is implemented by the experimenters being present in the same room and listening/seeing searching queries from users at a short distance. Being present in the same room, experimenters were able to provide constant searching results and observed users' behaviour at real time. A web application was designed explicitly for this study for easy access from any device and later refined the interface on the go after conducting the study with participants.

Our primary focus is on how multi-modality plays a role in conversational search, especially when several modalities are being mixed to drive interactions in conversational search. We present an example of such multi-modal conversation. Users can start by speaking specific words, or sentences, for search queries (e.g., "Haka dance"). Experimenters can then return an image grid to the users (i.e., traditional Haka dance or a rugby team performing Haka dance), and reiterate with text-to-speech what has been searched (e.g., "here are some images about *Haka*"). After that, users see the images and touch specific images they are interested in (e.g., the rugby team). They can also speak about the reason they are interested in the images (e.g., saying "yes, that's the one"). Then, the experimenter can refine the search term based on different modalities on the go and delivered refined images back to the users (e.g., more images of the rugby team performing the haka), as well as asking the user if the response is satisfying, using text-to-speech.

3.2 Phase 2: Experimentation

3.2.1 Participants. Participants were eight adults with intellectual disability, between 32 and 65 years old, presenting with a range of moderate to more severe intellectual disabilities. The participants were selected by support workers at of a partnering disability support organisation, located in a country from the Global North,

based on their familiarity with tablets, phones, and voice interfaces. 2 members of their support networks took part in the study, to both support their participation and provide insights. While it may seem helpful to present a description of the medical condition of individuals, we believe this may be not reflect individuals' abilities, and may raise privacy concerns. Therefore, as the study is exploratory in nature, we prefer to focus on a broader range of people with ID and engage with them to understand how they wish to use a multi-model search system. All participants gave their signed consent prior participating the study, the Queensland University of Technology Human Research Ethics Committee (approval number 2000000213) approved all procedures. In both settings, there was a researcher observing, video recording interactions, and conducting contextual interviews, as well as one or two researcher(s) acting as an experimenter, located in the same room. The experimenter(s) could easily hear the participants and wrote observation notes upon conclusion of the trials. While the different individuals interacted with the application, they were asked about their motivation for searching and their satisfaction with the answers returned. The research team discussed results together by identifying major themes observed in fieldwork notes, the application's logs, and recorded discussions with participants.

3.2.2 Setting up the scene. We conducted the study in two observation settings. The first was conducted with two participants taking part individually. They are an older woman and a 27-year-old man, both verbal with prior experience with search systems, tablet devices and voice-activated interfaces. They could type and read text mainly unassisted. The second was collaborative search experience, and took place within the context of a science class attended by six participants and two support workers.

Individual Search: The conversational application was first explained to potential participants at the centre, and they were invited to volunteer to interact with the app. The participants were asked to sign a consent form specially designed in a clear format to explain the simulation in clear and easy to read language. The participants were able to search for whatever information need they wanted, as no particular prompt or task was provided. The participants sat next to another researcher who also provided support if requested, and the application was launched on a touch screen tablet which

displayed the 'User' page. On a close-by table, the Wizard of Oz experimenter was set up with a laptop connected to the 'Helper' page.

Collaborative Search: At the centre, science classes are held weekly. The recent science topic they explored was magnetism, so we used this as a prompt for the collaborative search setting. All six participants in the class, as well as the two support workers leading the class, agreed to take part in the study after being provided with appropriate information, verbally and in writing. The participants and tutors were seated in the room, while the 'User' screen of the application was displayed on an interactive whiteboard at the front to be viewed by all participants. Two experimenters were seated at the front of the room with two laptops connected to the 'Helper' page of the application, acted as the conversational agent. Having two decreased the waiting time experienced by the users.

4 PRELIMINARY RESULTS AND DISCUSSION

We present our findings across individual and group trials. From observation results, we then discussed the implication of multimodal search and context needs, and reflection.

4.1 Observations

4.1.1 Individual search. No prompt was given to an individual information retrieval session; therefore, the participants formulated their queries related to their interests and entertainment. For instance, the female participant asked the "system" varied topics, with a limited logical connection. She conducted most of her queries using the on-screen touch keyboard; however, this input was slow and caused spelling errors. Her usage of text input may be due to the option of voice input not being clear enough to her, as when reminded she could speak to the "system", "You can also talk to me instead of typing if it's easier!", she would switch to using voice input. The other male participant preferred to voice input for his queries exclusively and spoke clearly and slowly with specific questions such as, "Show me a picture of the bush-walk." Regarding the usage of touch and images by participants, the female participant would often tap the images returned in the image grid, which would be transmitted to the Helper. For instance, when she queried to see photos of the Haka dance with her voice, an image grid was sent containing the Haka performed in different ways, ranging from modern to more traditional. This allowed her to refine her search query further, as she exclaimed "Look at that one!" as she tapped on the more traditional Haka performance. The Helper was then able to refine the search term from "Haka performance" to "Traditional Haka performance" and return more images of her query. When receiving the new images, she stated, "That's the one I want!", confirming the success of the refinement of her search. In contrast, the male participant did not interact with the touch modalities of the application. This may be due to the participant's vision issues, as to view the images, he had to put his face close to the screen, making pointing and touching less of a natural response.

4.1.2 Collaborative search. As the prompt of the recent science topic of magnets was given, participants and support workers developed queries surrounding magnets and how they work. Due to the User page being displayed on a big screen for everyone, the modality of touch could not be used in this setting. Queries were all

given by voice, typically in a question format such as "Are magnets in fridges?" In a collaborative environment with support workers and people with intellectual disability, queries were often formed with either assistance from support workers or their peers. Support workers would sometimes suggest queries to participants, which the participant could then repeat out loud themselves. Participants would also come up with their own queries inspired by their science classes, and either ask them out loud to the "system" or say the query to a support worker first to gain assistance on formulating the sentence. The ability of the "system" to constantly 'hear' queries allowed for support workers to speak with participants, as the support workers would sometimes speak the words required for the query one by one, allowing the participant to repeat after them. One user particularly benefited from this support, as she had a stutter, making it difficult for her to vocalise her information query. She was able to vocalise her query herself with encouragement from the support staff, however her query took a long time for her to say due to her stutter. At first, she thought her query had failed when the "system" did not immediately respond and was upset. The support worker reassured her that the "system" had heard her, and when it responded to her query, she was visibly excited and happy. Participant's own queries were also sometimes inspired by their peers, creating a logical connection between questions within a topic. An example of this was someone asking, "Do fridges have magnets?". Hearing this query and answer inspired a different user to ask, "Do dishwashers have magnets?", due to the connection between kitchen appliances and magnets.

4.2 Conversational search with prototype and its context

4.2.1 Context for conversational search. Context was used to guide the conversational search between participants and the "system". An example of this occurring was when the "system" was queried on what is attracted to magnets. As the Wizard of Oz Helper was aware that the participants had recently done an experiment using paper clips with magnets, they used this context to answer the question with "Iron, Cobalt and nickel and steel are attracted to magnets; That's why the paper clips were used in your experiment," and sent a photo of a typical paper clip magnet experiment. This image was recognised by the participants, with the support workers pointing out that it was the same as their previously conducted experiment. The "system" could also use the context of conversations occurring around it to aid in communication and inspire new information retrieval. In the individual setting, the male participant was asking the other researcher about which building she works at in the University. As the researcher was attempting to explain the location of the building, the "system" sent through a photo of the correct building, which aided in communicating to the participant.

4.2.2 User experience and satisfaction. Understanding participant experiences are valuable to gather their interests and aid the "system" to start/continue the conversation. This occurred in the collaborative setting when the question "Where do you find magnets?" was asked. The Wizard of Oz researcher did not understand this question, so asked a query refinement question of "Would you like to buy some magnets?". Multiple users replied with the natural language "No", and the support worker further helped refine the

user's query to, "Where do you find them in the house?" By being able to ask refinement questions, the query was better understood, and the correct information was returned to the users. The "system" could also use initiative to inspire new conversations around the same topic, much like a real conversation. Similar to conversational search group, this also occurred in the individual scenario when the information topic was changed from bush-walking to wildlife through questioning by the "system". After sending requested photos of bush-walks, the "system" took initiative to ask, "What is your favourite part of the bush-walk". When the user replied, "Encountering the amazing variety of koalas", the "system" sent photos and information about koalas, connecting the previous queries to a new topic of conversation. Feedback was given and observed on the user experience and satisfaction from both users with intellectual disability and support workers. In all simulation settings, it was observed that the participants did not notice the application being a 'Wizard of Oz', as when interacting with the application, participants only addressed the 'User' screen. This was further confirmed by a male participant asking if he could download the app to his iPad, not understanding that the "system" was not real. Therefore, the user experience for people with intellectual disability was very similar to interacting with a real smart conversational search system.

4.3 Discussion

This section discusses critical implications of using the "system" to understand how people with intellectual disability seek information through multi-modal conversational search, and highlights opportunities for further study.

4.3.1 Multi-modal Uses and Information Needs. Off-the-shelf conversational search system generally provide limited combinations of modalities, thus not fulfilling their potential to support people with intellectual disability to engage with online information in their own terms. In our study, users connect images, touch, and voice together to communicate their information needs. Images and voice were found to attract participants' attention and drive the conversation between the "system" and users. However, we observed a limited usage of touch in one of our individual search setting as the participant had to bring the screen very close to his eyes in order to see the images, while touch requires further distance to the screen. In the collaborative search setting, the ability to touch was not used as the big screen was not able to touch. It was possible for participants to stand up to touch the images, or to point to the images from across the room, however none of them used this modality. Thus when the touch modality is considered in multi-modal conversational search, it should be designed to accommodate the various context in which this interaction can take place, and for providing alternative ways to achieve the same level of interaction. Interestingly, using a collaborative setting could inspire and encourage information seeking for participants with intellectual disability, as new queries were sometimes inspired by hearing their peers' own questions. An example of this was someone asking, "Do fridges have magnets?". Hearing this query and answer inspired a different user to ask, "Do dishwashers have magnets?", due to the connection between kitchen appliances and magnets. This opportunity for collaboration via voice queries is not possible with current

search systems [7], indicating that conversational search systems could be particularly useful for collaborative learning.

4.3.2 Context contributing to conversational search. A previous study [1] noted that people with intellectual disability's mental model of VAs was that of a person, which was also seen when interacting with the conversational system via voice. Therefore, the ability of a conversational system to maintain context was observed to be beneficial, as users would often use natural language queries that would not be understood by current VAs. For example, when the "system" suggested "You can take Haka Classes in Brisbane" the participant asked the follow-up question: "How can I do that?", using natural language with a reference to the previous statement (i.e. an anaphora). With the search queries asked by people with intellectual disability, the "system" could also use the context of the users' abilities and experiences, their recent activities at the centre, and their recent queries to improve the answer returned. Analysis here is not able to provide insights into whether the additional context improved participants' understanding of the information presented. However, this use of familiar context was noticed by the participants.

4.3.3 Providing Independence and Support. Most participants could independently use this "system" using their preferred modality of input. Users with minor speech impediment experienced increased independence compared to current VAs, as this "system" did not have the typical barriers of misunderstanding them or timing out before their query was finished [1]. For participants who required more support, the conversational "system" was seen to greatly support the 'Seek Assistance' stage of information retrieval [16]. By allowing support workers and users to speak together, the "system" allowed users with severe speech impediments to voice their query themselves, which is not possible with current voice search technology [1]. This highlighted the potential of conversational search to support users who could not typically use voice search or struggled to formulate queries themselves.

4.3.4 Reflection. Our study built an accessible prototype that could be adapted to participants' interest and motives, as we have more control over what modalities are in need, what context are useful for conversational search, and how we set up the prototype for gathering participants' data. Although this is a preliminary study, and does not present an in-depth analysis of the data, we explored the use of conversational search from different angles, ranging from user experiences to surrounding context. Users provided valuable feedback when interacting with the "system", however, there are some recommendations that researchers should consider upon expanding upon this study:

- Presenting different type of modalities. We presented three type of modalities: images, touch and voice; more trials and interactions with the "system" would support an understanding of the type of modalities that are in most needed and helpful.
- Reducing the "system"'s response time. This could be achieved by introducing predefined responses. The experimenter could be trained prior conducting the study, and two experimenters

could be helpful for reducing the waiting time for participants. Additionally, a multi-modal representation of the system working towards a response, such as an animation or spoken statement about the process, potentially including a repetition of the query, could make the time experimenters need to provide responses more natural.

- Setting up the scene is very helpful to conduct the study as it encourage participants to ask questions and engage with the conversational search. It is however important to recognise the biases that prompting can trigger.

This study was exploratory in nature, however it provided some useful insight for future work that can examine more closely the nature of conversations and interactions. This will help us to shape our future study while maximising the participation of users with intellectual disability in future iterations of intelligent designs.

5 CONCLUSION AND FUTURE WORK

The main goal of the current study was to understand how future conversational multi-modal systems can support people with intellectual disability seek information online. Although the application developed was an early stage of prototype, participants had positive experiences when interacting with the “system”. Throughout the study, we believe different type of modalities (i.e., images, touch and voice) helped users to search for relevant information. Setting the scene and context are both useful for the study to run and gather more understating into individual’s needs/motives. Interestingly, conversational systems were also seen to better support collaborative information retrieval, which had the potential to encourage people with intellectual disability to explore new topics and develop new information queries.

A future step of this study will be to conduct more trials, and improve the processes of setting-up the prototype based on existing recommendations. For future studies, the current application wait times experienced by users can be improved by improving the experience for the ‘Helper’ by adding response suggestions and an improved user interface. Different information retrieval prompts which are more relevant to images could also be examined, as some of the currently examined queries did not support image information well. Furthermore, the impact of including context when explaining information to people with intellectual disability could be examined, to determine if it has a substantial impact on understanding. The potential of collaborative search with conversational systems also requires further exploration and experimentation, as this was observed to have positive impacts on the search process for people with intellectual disability.

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